

Multi Frame Set Output Mode

The data except this specification conform to those of each product.

1. Product type

IMX530 series

IMX530-AAMJ-C, IMX530-AAQJ-C, IMX531-AAMJ-C, IMX531-AAQJ-C, IMX532-AAMJ-C, IMX532-AAQJ-C

IMX535 series

IMX535-AAMJ-C, IMX535-AAQJ-C, IMX536-AAMJ-C, IMX536-AAQJ-C, IMX537-AAMJ-C, IMX537-AAQJ-C

IMX540 series

IMX540-AAMJ-C, IMX540-AAQJ-C, IMX541-AAMJ-C, IMX541-AAQJ-C, IMX542-AAMJ-C, IMX542-AAQJ-C

IMX545 series

IMX545-AAMJ-C, IMX545-AAQJ-C, IMX546-AAMJ-C, IMX546-AAQJ-C, IMX547-AAMJ-C, IMX547-AAQJ-C

IMX548 series

IMX548-AAMJ-C, IMX548-AAQJ-C

IMX565 series

IMX565-AAMJ-C, IMX565-AAQJ-C, IMX566-AAMJ-C, IMX566-AAQJ-C, IMX567-AAMJ-C, IMX567-AAQJ-C

IMX568 series

IMX568-AAMJ-C, IMX568-AAQJ-C

2. Description

The function to control the gain and the exposure period according to each frame is described in this document.

Multi Frame Set Output is became possible the sequence control by setting the gain and the exposure period of each frame, assuming 2 frames or 4 frames to be one-set.

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Application circuits shown, if any, are typical examples illustrating the operation of the devices. Sony Semiconductor Solutions Corporation cannot assume responsibility for any problems arising out of the use of these circuits.

3. Features

◆ Multi frame set output mode

2 frame set (Multi Frame Set Output mode 2 frame)

4 frame set (Multi Frame Set Output mode 4 frame)

◆ Frame identifier

Frame information line (In case of SLVS output)

Embedded data line (In case of SLVS – EC output or CSI-2 serial data output)

◆ Support mode

Normal mode (XVS / XHS sync)

* Trigger mode is not supported.

◆ Support readout drive mode

All-pixel scan mode

◆ The monitor output for Exposure period

◆ Support interface

(IMX530 series / IMX535 series)

Interface
SLVS – EC output (1 lane / 2 lane / 4 lane / 8 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX540 series / IMX545 series)

Interface
SLVS – EC output (1 lane / 2 lane, 8 bit / 10 bit / 12 bit output)
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX548 series)

Interface
SLVS output (2 ch / 4 ch / 8 ch, 8 bit / 10 bit / 12 bit output)

(IMX565 series / IMX568 series)

Interface
CSI-2 serial data output (2 Lane / 4 Lane, 8 bit / 10 bit / 12 bit output)

◆ Support each frame gain adjust function

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Internal use only

4. Register Map

The register map for Multi Frame Set Output mode is shown below.
Please refer to the product specification for register setup other than those list.

Registers corresponding to Chip ID = 04h in Write mode. (I²C:32**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
08h	3208h	0	WDSEL [1:0]	WD frame number setting	0h	00h	V
		1		0d: Normal mode, 1d: Multi Frame Set Output mode 2 frame 2d: Setting prohibited 3d: Multi Frame Set Output mode 4 frame			
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—
40h	3240h	0	SHS1 [23:0]	LSB	000048h*1	000070h*2	V
		1					
		2					
		3					
		4					
		5					
		6					
		7					
41h	3241h	0		Storage time adjustment			
		1		Designated in line unit			
		2					
		3					
		4		In Multi Frame Set Output mode, designated for FID: 1.			
		5		(2 Frame and 4 Frame commonness)			
		6					
		7					
42h	3242h	0					
		1					
		2					
		3					
		4					
		5					
		6					
		7		MSB			

Address		bit	Register Name	Description	Default value after reset				Reflection timing									
4-wire	I ² C				By register		By address											
44h	3244h	0	SHS2 [23:0]	LSB	000048h ^{*1}	000070h ^{*2}	48h ^{*1}	70h ^{*2}	V									
		1																
		2																
		3																
		4																
		5																
		6																
		7																
45h	3245h	0		Storage time adjustment			In Multi Frame Set Output mode 2 Frame, designated for FID: 0. In Multi Frame Set Output mode 4 Frame, designated for FID: 2.	000048h ^{*1}		000070h ^{*2}	00h	V						
		1		Designated in line unit														
		2																
		3																
		4																
		5																
		6																
		7																
46h	3246h	0		SHS2 [23:0]							In Multi Frame Set Output mode 2 Frame, designated for FID: 0. In Multi Frame Set Output mode 4 Frame, designated for FID: 2.		000048h ^{*1}	000070h ^{*2}	00h	V		
		1																
		2																
		3																
		4																
		5																
		6																
		7																
48h	3248h	0	SHS3 [23:0]		LSB	000048h ^{*1}			000070h ^{*2}						48h ^{*1}		70h ^{*2}	V
		1																
		2																
		3																
		4																
		5																
		6																
		7																
49h	3249h	0			Storage time adjustment		In Multi Frame Set Output mode 4 Frame, designated for FID: 3	000048h ^{*1}		000070h ^{*2}		00h			V			
		1			Designated in line unit													
		2																
		3																
		4																
		5																
		6																
		7																
4Ah	324Ah	0		SHS3 [23:0]	In Multi Frame Set Output mode 4 Frame, designated for FID: 3						000048h ^{*1}	000070h ^{*2}	00h	V				
		1																
		2																
		3																
		4																
		5																
		6																
		7																
		0	SHS3 [23:0]			In Multi Frame Set Output mode 4 Frame, designated for FID: 3			000048h ^{*1}				000070h ^{*2}			00h	V	
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		5																
		6																
		7																
		0					SHS3 [23:0]	In Multi Frame Set Output mode 4 Frame, designated for FID: 3		000048h ^{*1}					000070h ^{*2}	00h		V
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Address		bit	Register Name	Description	Default value after reset				Reflection timing			
4-wire	I ² C				By register		By address					
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		4										
		5										
		6										
		7										
4Dh	324Dh	0		Storage time adjustment Designated in line unit			000048h ^{*1}	000070h ^{*2}		00h		
		1										
		2										
		3										
		4										
		5										
		6										
		7										
4Eh	324Eh	0		In Multi Frame Set Output mode 4 Frame, designated for FID: 0			000048h ^{*1}	000070h ^{*2}		00h		
		1										
		2										
		3										
		4										
		5										
		6										
		7										
				MSB								

^{*1} IMX530 series and IMX540 series.

^{*2} IMX535 series, IMX545 series, IMX548 series, IMX565 series and IMX568 series.

Registers corresponding to Chip ID = 06h in Write mode. (I²C:34**h)

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
36h	3436h	0	TOUT0SEL [2:0]	TOUT0 pin setting	0h	00h	S
		1		0h : Low fixed			
		2		1h : Exposure period monitoring			
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6		Fixed to 0	0		—
		7		Fixed to 0	0		—

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Registers corresponding to Chip ID = 07h in Write mode. (I²C:35**h)

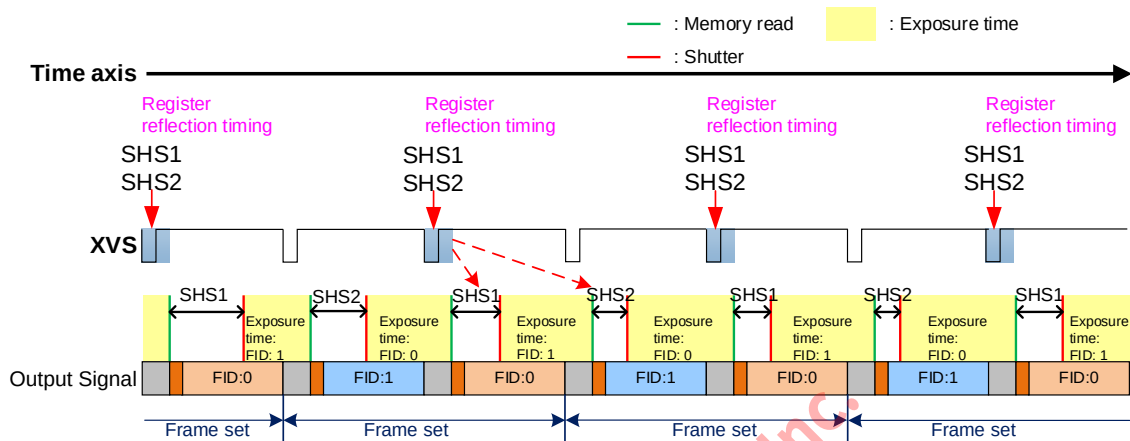
Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
00h	3500h	0		Fixed to 1	1	01h	—
		1		Fixed to 0	0		—
		2		Fixed to 0	0		—
		3		Fixed to 0	0		—
		4		Fixed to 0	0		—
		5		Fixed to 0	0		—
		6	WD_GAIN_MODE [0]	Setting of Multi Frame Set Output mode gain function 0 : Normal mode 1 : Each frame gain mode	0		S
		7		Fixed to 0	0	—	
14h	3514h	0	GAIN [8:0]	LSB	000h	00h	V
		1		Setting of gain adjustment Effective at "WD_GAIN_MODE = 0" Designated for FID: 0 to 3 identical gain			
		2					
		3					
		4					
		5		Effective at "WD_GAIN_MODE = 1" Designated for FID: 0			
		6					
		7					
0	MSB						
15h	3515h	1		0	00h	—	
		2		0		—	
		3		0		—	
		4		0		—	
		5		0		—	
		6		0		—	
		7		0		—	
		16h	3516h	0		GAIN1 [8:0]	LSB
1	Setting of gain adjustment Effective at "WD_GAIN_MODE = 1" Designated for FID: 1						
2							
3							
4							
5							
6							
7							
0		MSB					
17h	3517h	1		0	00h	—	
		2		0		—	
		3		0		—	
		4		0		—	
		5		0		—	
		6		0		—	
		7		0		—	

Address		bit	Register Name	Description	Default value after reset		Reflection timing
4-wire	I ² C				By register	By address	
18h	3518h	0	GAIN2 [8:0]	LSB	000h	00h	V
		1					
		2					
		3					
		4		Setting of gain adjustment Effective at "WD_GAIN_MODE = 1" Designated for FID: 2			
		5					
		6					
		7					
19h	3519h	0		MSB	0	00h	—
		1					
		2					
		3					
		4					
		5					
		6					
		7					
1Ah	351Ah	0	GAIN3 [8:0]	LSB	000h	00h	V
		1					
		2					
		3					
		4		Setting of gain adjustment Effective at "WD_GAIN_MODE = 1" Designated for FID: 3			
		5					
		6					
		7					
1Bh	351Bh	0		MSB	0	00h	—
		1					
		2					
		3					
		4					
		5					
		6					
		7					

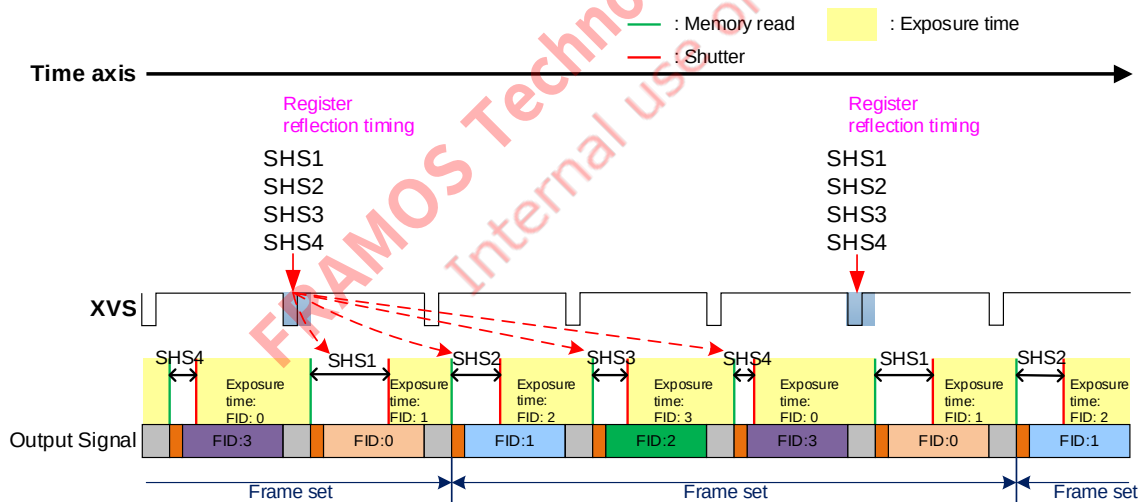
5. Reflection Timing for Exposure period

In Multi Frame Set Output mode, the setting of register is reflected only if the Frame ID is "0d". (About the FID, refer to next page.) Set the shutter settings in such a value that the magnitude relation is satisfied.

- ◆ 2 frame set: SHS1 > SHS2
- ◆ 4 frame set: SHS1 > SHS2 > SHS3 > SHS4



Timing of register setting for Multi Frame Set Output mode 2 frames



Timing of register setting for Multi Frame Set Output mode 4 frames

The exposure time of each frame is according to value the following register.

Frame ID	Multi Frame Set Output mode 2 frames [H]	Multi Frame Set Output mode 4 frames [H]
0	The number of line of 1 frame — (SHS2) + t_{OFFSET}^*	The number of line of 1 frame — (SHS4) + t_{OFFSET}^*
1	The number of line of 1 frame — (SHS1) + t_{OFFSET}^*	The number of line of 1 frame — (SHS1) + t_{OFFSET}^*
2	—	The number of line of 1 frame — (SHS2) + t_{OFFSET}^*
3	—	The number of line of 1 frame — (SHS3) + t_{OFFSET}^*

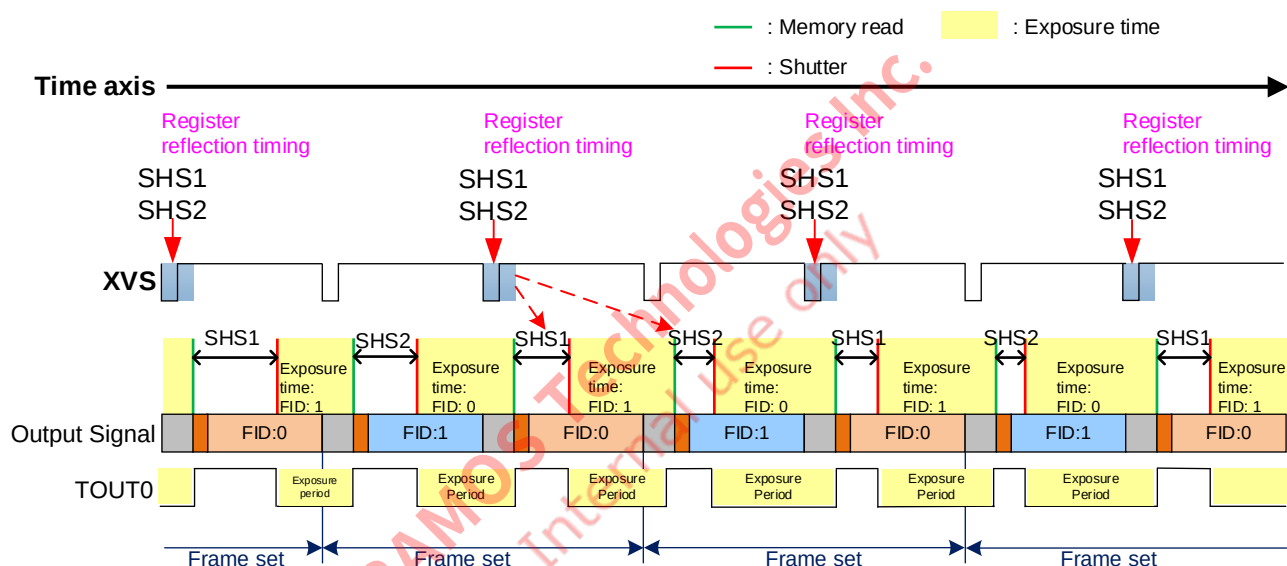
* Please refer to data sheet for t_{OFFSET}

6. The monitor output for Exposure period

As for this sensor, TOUT0 can output the status of the exposure control signal in Multi Frame Set Output mode.

6.1. List of registers

Register	Register details			Initial value	Description
	Chip-ID	Address (): I2C	bit		
TOUT0SEL [2:0]	06h	36h (3436h)	[2:0]	0h	TOUT0 pin setting 0h : Low fixed 1h : Exposure period monitoring



Exposure period monitor for Multi Frame Set Output mode 2 frames

7. Each frame Gain mode

The sensor becomes Multi Frame Set Output mode that the exposure times of each frame are controlled by setting register WDSEL. Then, set to each frame is available by setting 1h to register WD_GAIN_MODE. Thereby the gain settings set each frame, analog and digital individually. Setting step is 0.1 dB.

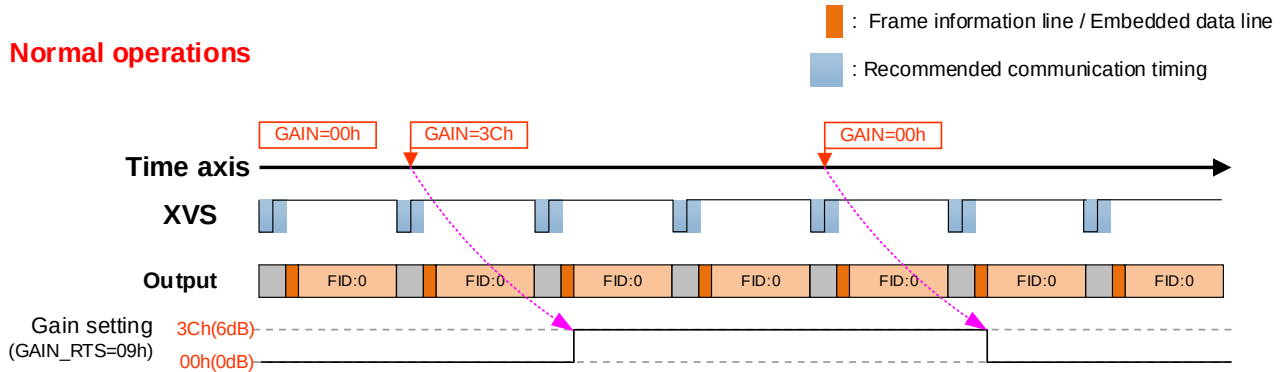
7.1. List of setting registers

Register details			Register	Description	Normal mode	Multi Frame Set Output mode	
Chip-ID	Address (): I2C	bit				2 frames	4 frames
04h	08h (3208h)	[1:0]	WDSEL [1:0]	In Multi Frame Set Output mode, set number of operation frames.	0h	1h	3h
07h	00h (3500h)	[6]	WD_GAIN_MODE [0]	Individually gain control	Fixed to "0h"	Disable: 0h Enable: 1h	Disable: 0h Enable: 1h
	14h (3514h)	[7:0]	GAIN [8:0]	Gain setting	Gain setting 0 to 48 dB	Frame ID: 0h Gain setting 0 to 48 dB	Frame ID: 0h Gain setting 0 to 48 dB
	15h (3515h)	[0]					
	16h (3516h)	[7:0]	GAIN1 [8:0]	Gain Setting	Fixed to 000h	In WD_GAIN_MODE = 1h, Frame ID: 1h Gain setting 0 to 48 dB	In WD_GAIN_MODE = 1h, Frame ID: 1h Gain setting 0 to 48 dB
	17h (3517h)	[0]					
	18h (3518h)	[7:0]	GAIN2 [8:0]	Gain setting	Fixed to 000h	Fixed to 000h	In WD_GAIN_MODE = 1h, Frame ID: 2h Gain setting 0 to 48 dB
	19h (3519h)	[0]					
	1Ah (351Ah)	[7:0]	GAIN3 [8:0]	Gain Setting	Fixed to 000h	Fixed to 000h	In WD_GAIN_MODE = 1h, Frame ID: 3h Gain setting 0 to 48 dB
	1Bh (351Bh)	[0]					

8. Gain reflection timing

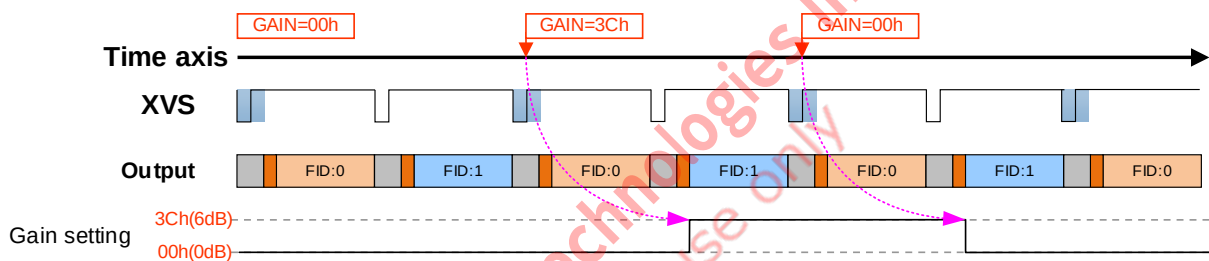
In Multi Frame Set Output mode, the reflection timing of gain setting is reflected collectively to the unit of frameset. The gain reflection timing is shown figure below. (In Multi Frame Set Output mode, a special setting is unnecessary.)

Normal operations



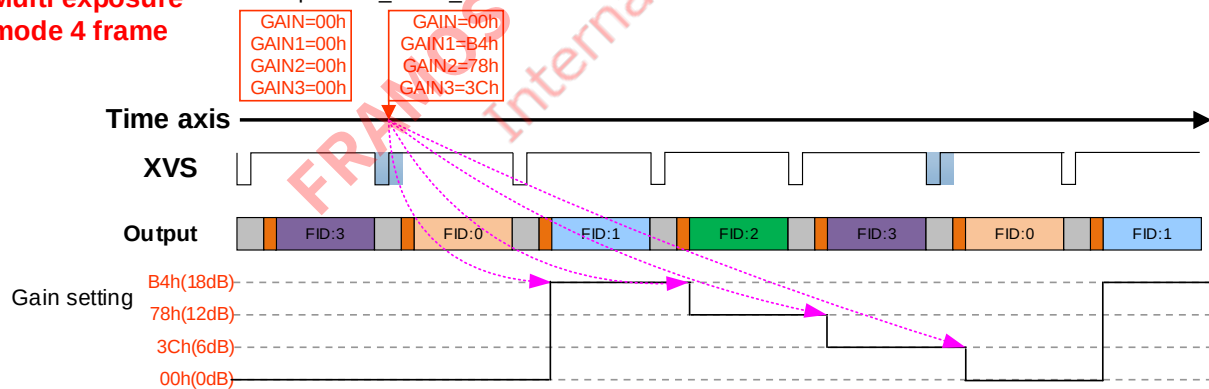
Multi exposure mode 2 frame

Example: WD_GAIN_MODE = 0



Multi exposure mode 4 frame

Example: WD_GAIN_MODE = 1



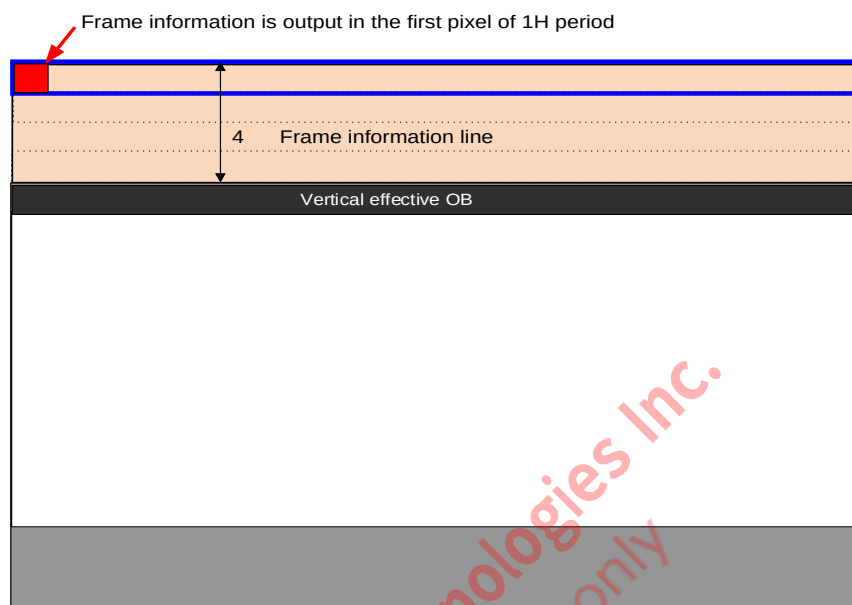
Gain Reflection Timing

9. Frame identification method

The identification signal of each frame is Frame ID, and the unit of Multi Frame Set Output mode is Frame set.

9.1. Frame information line

This product uses Frame information line of SLVS to output Frame ID and Frame set. It is added to the output signal of the 1st pixel of the blue frame in the figure below.



Frame information line output position

9.2. Output format (Frame information line)

The output format of Frame information line is shown below.

Output	8bit
[7]	1
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

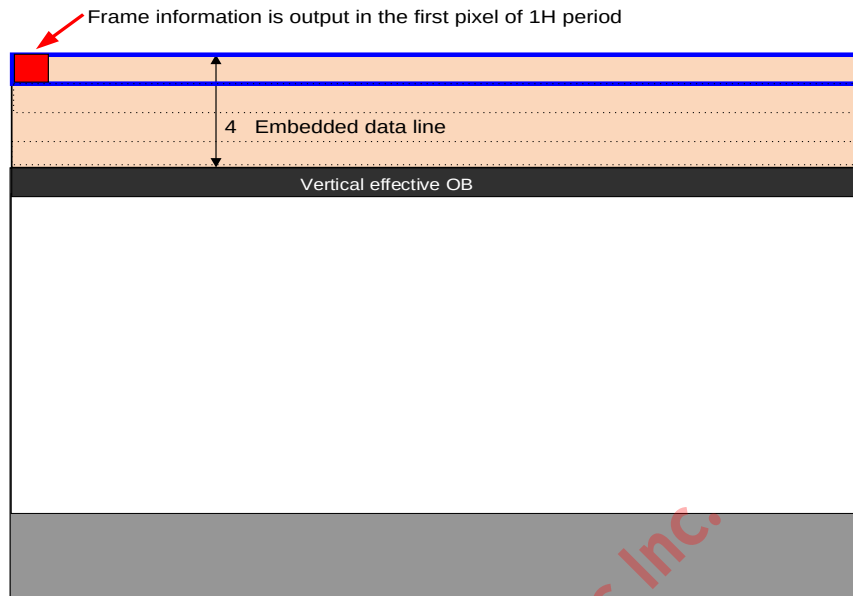
Output	10bit
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	12bit
[11]	0
[10]	0
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output format (Frame information line)

9.3. Embedded data line

This product uses Embedded data line of SLVS – EC or CSI-2 to output Frame ID and Frame set. It is added to the output signal of the 1st pixel of the blue frame in the figure below.



Embedded data line output position

9.4. Output format (Embedded data line)

The output format of embedded data line is shown below.

(SLVS-EC)

Output	8bit
[7]	1
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	10bit
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	12bit
[11]	0
[10]	0
[9]	1
[8]	Frameset
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output format (Embedded data line (SLVS-EC))

(CSI-2)

Output	8bit
[7]	Frame ID [1]
[6]	Frame ID [0]
[5]	0
[4]	0
[3]	0
[2]	Frameset
[1]	Frame ID [1]
[0]	Frame ID [0]

Output	10bit
[9]	Frame ID [1]
[8]	Frame ID [0]
[7]	0
[6]	0
[5]	0
[4]	Frameset
[3]	Frame ID [1]
[2]	Frame ID [0]
[1]	0
[0]	1

Output	12bit
[11]	Frame ID [1]
[10]	Frame ID [0]
[9]	0
[8]	0
[7]	0
[6]	Frameset
[5]	Frame ID [1]
[4]	Frame ID [0]
[3]	0
[2]	1
[1]	0
[0]	1

Output format (Embedded data line (CSI-2))

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10. Mode Transition

The transition related to Multi Frame Set Output mode is shown below. If perform the transition which is not described in table below, set via sensor standby.

10.1. Mode Transition

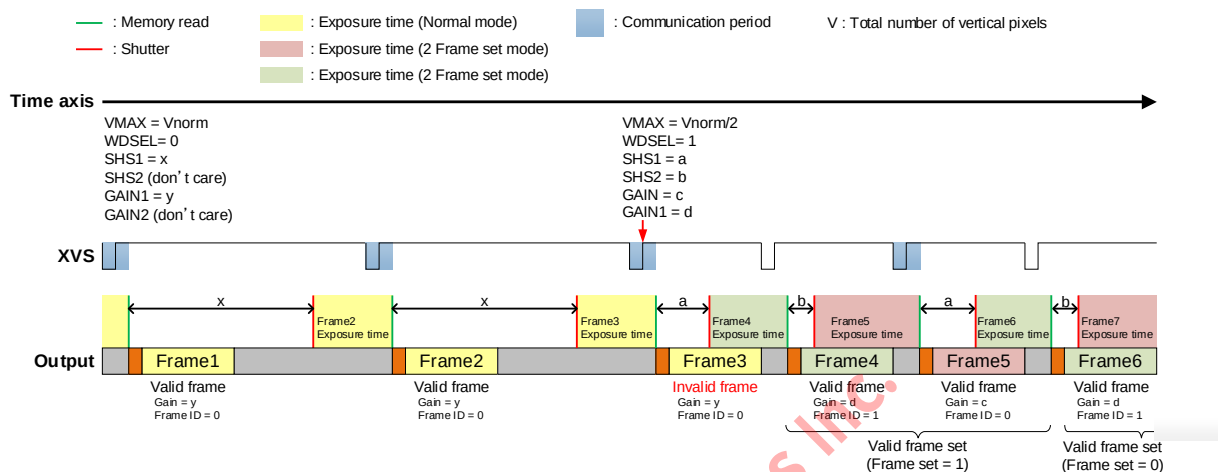
Mode Transition			Remarks
Normal mode	⇔	Multi Frame Set Output (2 Frame)	There comes out invalid frames. Changing the register value of FREQ, HMAX is prohibited. In case of setting as the same frame rate, set the 1 frame period of normal mode which to be same as 1 frame set period of Multi Frame Set Output mode.
Normal mode	⇔	Multi Frame Set Output (4 Frame)	There comes out invalid frames. Changing the register value of FREQ, HMAX is prohibited. In case of setting as the same frame rate, set the 1 frame period of normal mode which to be same as 1 frame set period of Multi Frame Set Output mode.
Multi Frame Set Output (2 Frame)	⇔	Multi Frame Set Output (4 Frame)	Set via the sensor standby.

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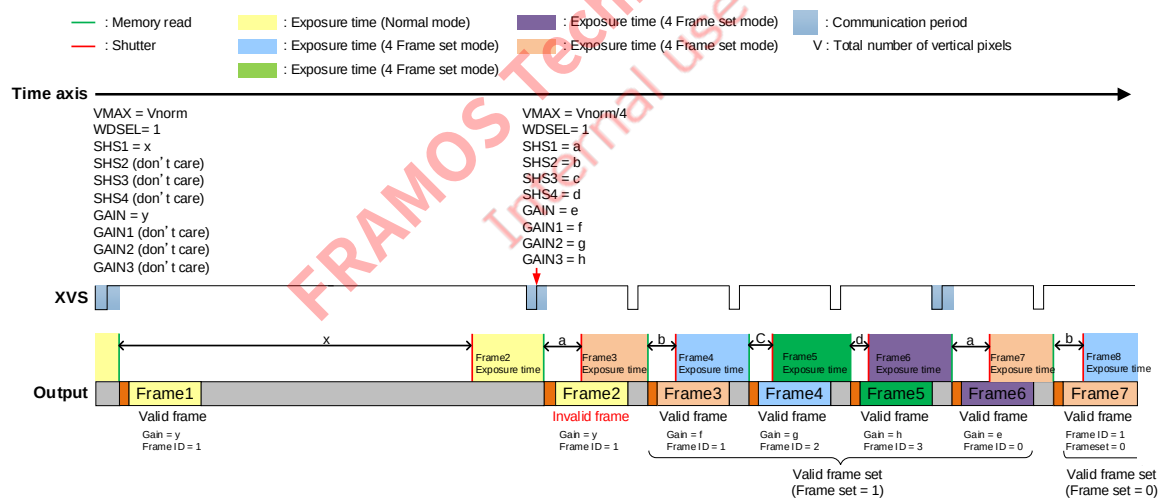
As one example, in case of setting to the same frame rate before the mode transition and after the mode transition, it is shown as follows.

◆ Transition from normal to Multi Frame Set Output mode

The transition setting frame becomes invalid.



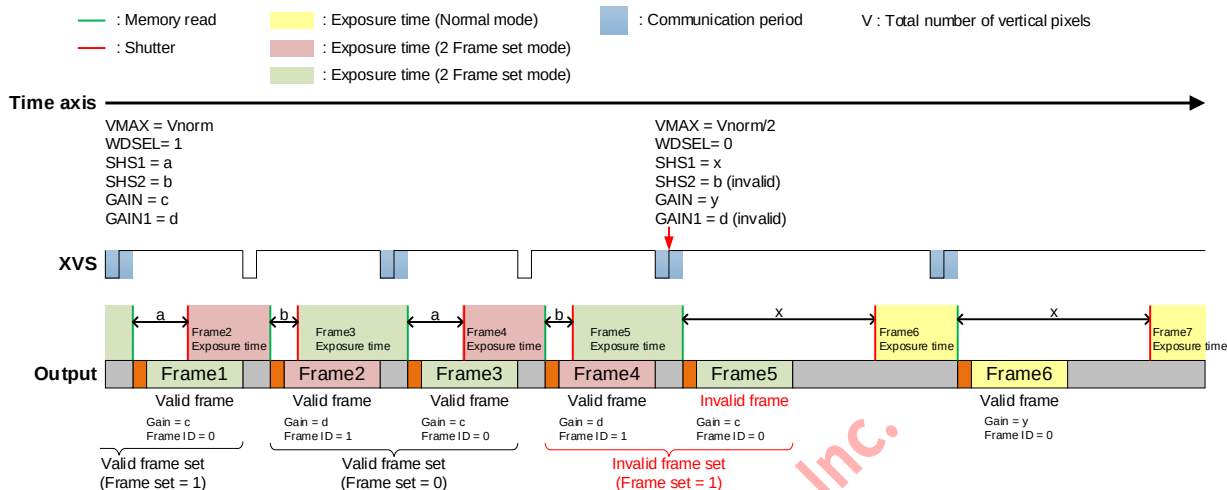
Transition to Multi Frame Set Output mode (2 frames) from Normal mode (Each frame gain mode)



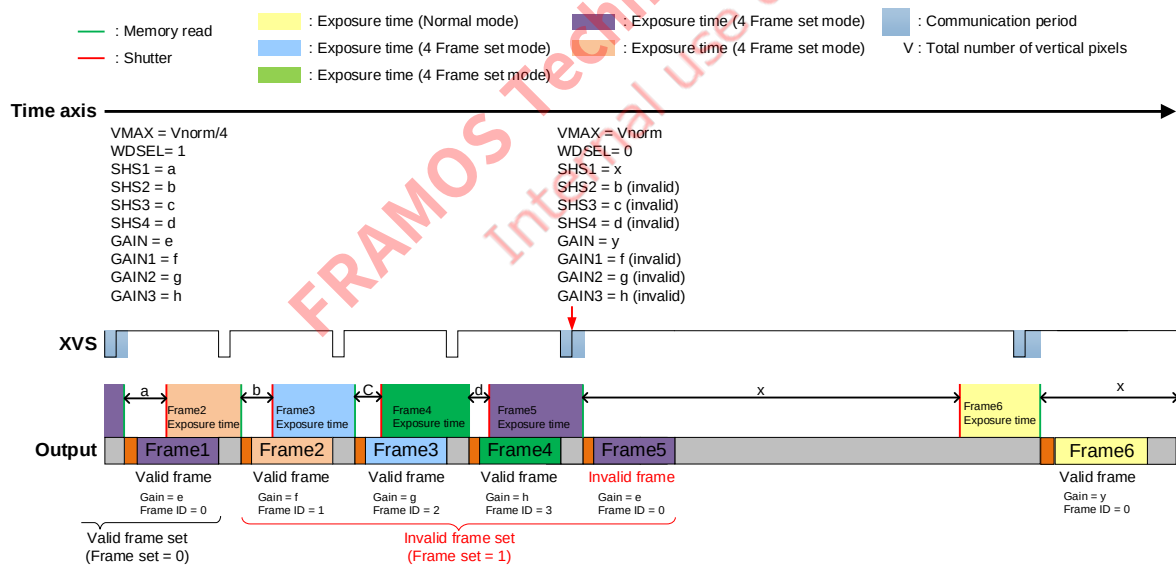
Transition to Multi Frame Set Output mode (4 frames) from Normal mode (Each frame gain mode)

◆ Transition from Multi Frame Set Output mode to normal mode

The transition setting frame becomes invalid. Because the transition setting frame is long term exposure signal of Multi Frame Set Output mode frameset, the last signal set become invalid.



Transition to Normal mode from Multi Frame Set Output mode (2 frames) (Each frame gain mode)



Transition to Normal mode from Multi Frame Set Output mode (4 frames) (Each frame gain mode)

11. Notes On Handling

Notes On Handling of the Multi Frame ROI mode.

- ◆ Multi Frame ROI mode and Dual ADC (without combination / with built-in combination) come by combination drive.
(only IMX530 series and IMX535series)
- ◆ The drive with the trigger mode is non-support.
- ◆ Multi Frame ROI mode and Short interval shutter function do not come by combination drive.
- ◆ Multi Frame ROI mode and Self Trigger do not come by combination drive.

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Revision History

Version	Date	Page	Remarks
Rev. 0.1	2019 / 4 / 26	–	First Edition
Rev.0.2	2019 / 6 / 28	1, 2, 9, 10	Correction : Function name Exposure time ⇒ Exposure period
		18	Correction : Function name Dual ADC (without combination / with built-in combination) Short interval shutter function
Rev.1.0	2020 / 6 / 2	–	First Edition (Official version of IMX530 series / IMX540 series)
		1	Added Chapter "Product type" Added SLVS 2 ch output in support interface Added the restriction in support interface
		12	Update the figure of Gain Reflection Timing
Rev.2.0	2020 / 12 / 17	–	Added: IMX535 series / IMX545 series / IMX565 series
		2, 16	Added: CSI-2 serial data output
		4, 5, 6	Added: Default Value after reset of IMX535 series, IMX545 series, IMX565 series
Rev.3.0	2021 / 3 / 4	1, 2, 6	Added: IMX548 series / IMX568 series
		20	Added: Definition of series for Dual ADC combination drive